

MINA ELZEK

Forward Deployed Engineer | AI Solutions Engineer | Technical Consultant

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PROFESSIONAL SUMMARY

Ships production AI and automation end-to-end. Portfolio: Undercut (playundercut.com)—live F1 prediction platform (Django 4.2 LTS, React 18/TypeScript, Railway + Cloudflare, CI/CD, Playwright); clinical CV raising polyp detection 50%→94%; operator-facing PLC/HMI (60+ instruments, 100+ limits, 40+ spec gaps); ProcessMind industrial RAG; CopticChat local RAG (~80k passages, 126 PDFs). LLM evaluation at Outlier. Open to FDE / AI Solutions / Technical Consultant roles—GTA hybrid/on-site.

CORE COMPETENCIES

Prototype → Production · System Integration · Production Inference (CV + RAG) · Local/On-Prem AI · Industrial PLC/HMI · Technical Discovery · CI/CD & E2E · Stakeholder/Operator Delivery

PROFESSIONAL EXPERIENCE

AI Prompt Engineer | Outlier AI · August 2020 – Present · Remote, Contract

- Remediated ~1,500–3,000 LLM outputs (multi-turn, tool-use, long-form) at 95%+ on-time; lifted rubric pass-rates ~20–35% via rewrites on factuality, constraints, and reasoning.
- Built evaluation frameworks and output taxonomies across reasoning, code, and creative tasks; fed quality signals into RLHF-style loops and production-oriented prompt optimization.

SELECTED PROJECTS

Undercut — Live F1 Prediction Platform | playundercut.com · July 2026 · End-to-end ownership

- Shipped production private F1 fantasy leagues: championship and race/sprint picks, bonus picks, invite-code leagues, Race Hub, live standings, and post-race receipts—2026 calendar via live OpenF1/Jolpica sync.
- Owned full stack and releases: Django 4.2 LTS + DRF (feature flags, 68 migrations, 60 test modules), React 18/TypeScript/Vite 6, PostgreSQL, Redis, Celery; Railway + Cloudflare; Stripe Plus; Playwright E2E; Capacitor iOS/Android.

Polyp Detection AI — Clinical Computer Vision | Endoch Medical · Real-time inference

- Raised polyp detection from 50% (physician visual baseline) to 94% model accuracy through iterative build with clinical stakeholders; real-time frame-level inference with validation UI.
- Integrated data→model→review: ETL on 3 datasets (~4,000–13,000 images each), preprocessing (artifacts, boxes/masks, normalization), and formal testing/validation reports.

PLC/HMI Process Automation — Food & Beverage | PID Engineering Inc · Operator-facing systems

- Delivered FactoryTalk View (SCADA/HMI) dashboards for dairy ops: 60+ instruments and 7 live KPIs (temperature, conductivity, flow, tank volume, pressure, step duration, concentration).
- Closed 40+ process-spec vs software gaps; implemented 100+ operational/safety limit alerts (DFO & CFIA context); Allen-Bradley ControlLogix/CompactLogix logic for mixproof sequencing, route control, and CIP.

ProcessMind AI — Industrial RAG | LangChain · ChromaDB · Ollama · Multi-agent

- Built industrial process copilot with RAG over maintenance manuals/SOPs (ChromaDB + LangChain), multi-agent diagnostics (LangGraph), and local LLM via Ollama; FastAPI + Next.js with human-in-the-loop explanations.

CopticChat — Local Domain RAG | 126 PDFs · ~80k passages · Next.js 15 · Ollama

- Local study RAG over 126 English library PDFs (~35k pages → ~80k sentence-aware passages); hybrid lexical + nomic-embed-text retrieval on SQLite FTS; claim-checked Ollama answers—citation-first, fail-closed when evidence is thin.

EDUCATION & CERTIFICATIONS

B.Eng, Computer Engineering with Honours — University of Wollongong · Jan 2020 – Apr 2023

PLC Technician II — George Brown College (Oct 2025) · Google Data Analytics Professional Certificate — Coursera (Jan 2026)

TECHNICAL SKILLS

Languages & Data: Python, TypeScript/JavaScript, SQL, MATLAB, Excel VBA, Git

AI / ML: TensorFlow, OpenCV, LangChain, LangGraph, ChromaDB, RAG, Ollama, Hybrid Retrieval, Prompt Engineering, LLM Evaluation

Full-Stack & Production: Django 4.2 LTS, DRF, React 18, Vite 6, Next.js 15, FastAPI, PostgreSQL, Redis, Celery, Docker, Railway, Cloudflare, CI/CD, Playwright, Capacitor, Stripe

Industrial: Allen-Bradley ControlLogix/CompactLogix, FactoryTalk View, Ladder/ST, HMI/SCADA, CIP, Process Monitoring

Delivery: Technical Discovery, Requirements Translation, Validation Reports, Stakeholder Collaboration, Hybrid/On-site GTA